

Paper 1, Theorems 1.2–1.3: Statement Amendment

Internal research note for the **ShenWork** formalization

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Abstract

The stability proof in Section 5 of Shen’s traveling-wave paper has three independent statement-level defects. The perturbed quadratic in (5.31) is positive at the unperturbed tail exponent, so the root comparison in (5.35) has the wrong direction; the weighted norm in (1.21) is written in laboratory rather than moving coordinates; and the exponential factor following (5.35) has the wrong sign. This note gives the direct amendment supported by the written proof, explains the corresponding change to the uniqueness theorem, and records the exact Lean certificates and remaining analytic boundary.

1 Source checked

The source examined is the original submission, not a repository paraphrase:

Wenxian Shen, *Existence, uniqueness, stability, and monotonicity of traveling waves for repulsion/attraction chemotaxis models with logistic type source*, arXiv:2605.04401v1, 6 May 2026.

The archived local copy is `paper1.pdf`. Theorem 1.2 claims weighted stability for every

$$\kappa < \eta < \frac{1}{1 + |\chi|^{1/6}},$$

where $\kappa = (c - \sqrt{c^2 - 4})/2$, and writes the weighted conclusion in laboratory coordinates.

2 The root comparison in (5.35)

The energy coefficient obtained in (5.31) is

$$q(\eta) = \eta^2 - (c - A)\eta + (1 + B), \tag{1}$$

where, in the notation of (5.32)–(5.33),

$$A = |\chi|^{1-3\sigma} D', \quad B = |\chi|^{1-3\sigma} D''.$$

The displayed formulas give $A \geq 0$ and, when $\chi \neq 0$, $B > 0$. The unperturbed tail exponent satisfies

$$\kappa^2 - c\kappa + 1 = 0.$$

Substitution into (1) therefore gives the exact identity

$$q(\kappa) = A\kappa + B > 0. \tag{2}$$

Consequently κ cannot lie between the two roots of q . Under the paper's speed inequality it lies strictly below the perturbed lower root:

$$\kappa < \kappa_{c,\chi,m,\alpha,\gamma}^-.$$

By continuity, (2) also produces weights strictly above κ for which $q(\eta) > 0$. Thus excluding only the endpoint $\eta = \kappa$ does not repair (5.35). The global energy estimate proves decay only above the perturbed lower root.

3 The coordinate mismatch in (1.21)

Section 5 changes to the moving coordinate $z = x - ct$ and estimates

$$E_{\text{move}}(t) = \int_{\mathbb{R}} e^{2\eta z} |u(t, z + ct) - U(z)|^2 dz. \quad (3)$$

The printed formula (1.21) instead uses

$$E_{\text{lab}}(t) = \int_{\mathbb{R}} e^{2\eta x} |u(t, x) - U(x - ct)|^2 dx. \quad (4)$$

The substitution $x = z + ct$ gives

$$E_{\text{lab}}(t) = e^{2\eta ct} E_{\text{move}}(t). \quad (5)$$

Hence decay of (3) does not imply decay of (4) without an additional rate stronger than $2\eta c$, which the proof does not supply. The natural correction is to use (3), equivalently the weight $e^{2\eta(x-ct)}$ in laboratory coordinates.

4 The sign after (5.35)

The paper defines the value $\lambda = q(\eta)$ to be negative and then writes an upper bound with $e^{-\lambda t}$, claiming that it tends to zero. For $\lambda < 0$, however, $-\lambda > 0$ and this factor grows. With the displayed definition the decaying factor is $e^{\lambda t}$. Equivalently, define the positive rate $\rho = -\lambda$ and write $e^{-\rho t}$.

5 Recommended amended statement

Let A_χ and B_χ be the limiting error budgets in (5.32)–(5.33), evaluated at the asymptotic bound M_χ , and set

$$\begin{aligned} \Delta &= (c - A_\chi)^2 - 4(1 + B_\chi), \\ \kappa^- &= \frac{(c - A_\chi) - \sqrt{\Delta}}{2}, \\ \kappa^+ &= \frac{(c - A_\chi) + \sqrt{\Delta}}{2}. \end{aligned}$$

Choose the speed threshold so that $\Delta > 0$ and

$$\kappa^- < \frac{1}{1 + |\chi|^{1/6}} < \kappa^+.$$

Recommended amendment to Theorem 1.2. Replace the condition $\kappa < \eta$ by

$$\kappa_{c,\chi,m,\alpha,\gamma}^- < \eta < \frac{1}{1 + |\chi|^{1/6}},$$

and replace (1.21) by

$$\lim_{t \rightarrow \infty} \int_{\mathbb{R}} e^{2\eta z} |u(t, z + ct; u_0) - U^*(z)|^2 dz = 0.$$

Retain the uniform conclusion (1.22), subject to the compactness and far-left rigidity argument in Step 4. Write the energy decay as $e^{\lambda t}$ for $\lambda < 0$, or as $e^{-\rho t}$ for $\rho = -\lambda > 0$.

Theorem 1.3 uses Theorem 1.2 with the second wave as initial data. Its common tail exponent must therefore leave room for the corrected stability weight: the hypothesis should require $\kappa_1 > \kappa^-$, not merely $\kappa_1 > \kappa$. The waves constructed in Theorem 1.1 satisfy a family of tail estimates, so such a rate can be selected once κ^- lies below the construction's tail cap.

6 What would be needed to retain the original interval

Equation (2) obstructs only the *global coefficient* estimate used in (5.31); it is not a counterexample to stability itself. The perturbation coefficients may decay in the right tail. Retaining every $\eta > \kappa$ would require a new spatially localized coercivity or weighted spectral argument. No such argument appears in Section 5. It would be a new theorem, not a formalization of the written proof.

7 Lean certificates and current boundary

The repository contains zero-sorry, standard-axiom certificates. The algebraic root certificates in `ShenWork/Paper1/Theorem12RootObstruction.lean` are

- `paper531_kappa_not_between_perturbed_roots;`
- `paper531_kappa_lt_rootMinus;`
- `paper531_positive_inside_stated_weight_window.`

The coordinate certificate `laboratoryWeightedL2Energy_eq_exp_mul_coMoving`, together with a non-vacuous coordinate witness, is in `ShenWork/Paper1/Theorem12CoordinateAudit.lean`. The two sign certificates `paper531_printed_decay_factor_tendsto_atTop` and `paper531_corrected_decay_factor_tendsto_zero` are in `ShenWork/Paper1/Theorem12RootObstruction.lean`.

`ShenWork/Paper1/Theorem12Corrected.lean` proves the scalar Gronwall and moving-frame localization wiring, but keeps the general-data whole-line PDE energy and compactness input explicit. The full amended stability theorem is therefore not yet claimed as proved. The corrected structural reduction for Theorem 1.3 is in `ShenWork/Paper1/Theorem13Corrected.lean`; its remaining analytic input is the same bounded-solution stability theorem.